

Lightweight Memory-Augmented PPO Agent Inspired by EMPO² for Exploration in Reinforcement Learning

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Abstract—Reinforcement Learning (RL) has achieved significant success in sequential decision-making problems. However, traditional RL algorithms often struggle in sparse-reward environments where effective exploration is required before useful rewards can be observed. Recent Agentic AI frameworks such as EMPO² introduce memory-augmented exploration mechanisms that enable agents to store, retrieve, and utilize past experiences to improve exploration efficiency.

This paper presents EMPO-Light, a lightweight implementation inspired by EMPO² that can be trained using limited computational resources available on Kaggle and Google Colab. The proposed framework combines Proximal Policy Optimization (PPO), an experience memory buffer, cosine-similarity-based memory retrieval, and a novelty-driven intrinsic reward mechanism.

Experiments were conducted on MiniGrid environments. PPO achieved an average reward of approximately 0.99 on MiniGrid-Empty-8x8-v0, demonstrating successful learning in a simple navigation environment. However, PPO failed to solve MiniGrid-DoorKey-8x8-v0 within 200,000 training timesteps, achieving an average reward of 0.0. The proposed EMPO-Light framework successfully stored 5,000 experiences, identified 320 novel states, and maintained a top retrieved experience reward of 63.9 through novelty-based exploration. Although the framework did not fully solve the DoorKey environment, it demonstrated the feasibility of memory-augmented exploration under limited computational resources.

Index Terms—Reinforcement Learning, PPO, Agentic AI, EMPO², Memory-Augmented Learning, Exploration, MiniGrid

I. INTRODUCTION

A. Background

Reinforcement Learning (RL) is a machine learning paradigm in which an agent learns to make decisions through interaction with an environment. At each timestep, the agent observes a state, performs an action, and receives a reward signal. The objective is to maximize cumulative rewards over time.

Recent advances in Agentic AI have extended traditional RL by incorporating reasoning, planning, memory, and long-term decision-making capabilities. Modern agentic systems aim to operate autonomously in complex environments while maintaining awareness of previous experiences.

One of the major challenges in RL is exploration. When rewards are sparse, agents may fail to discover rewarding states, resulting in poor learning performance. This issue becomes particularly significant in environments that require multi-step planning and long-term memory.

B. Problem Statement

Traditional PPO agents perform well in simple environments but often struggle in sparse-reward tasks requiring exploration and memory. MiniGrid-DoorKey-8x8-v0 is an example of such an environment, where the agent must: (1) locate a key, (2) pick up the key, (3) locate the locked door, (4) unlock the door, and (5) reach the goal state. Failure to discover any of these intermediate steps prevents the agent from receiving meaningful rewards.

C. Motivation

The recently proposed EMPO² framework introduces memory-driven exploration mechanisms designed to improve learning efficiency in sparse-reward environments. However, reproducing the complete EMPO² architecture requires substantial computational resources and advanced training infrastructure.

This paper investigates whether the core ideas of EMPO² can be approximated through a lightweight framework suitable for educational and research purposes.

D. Contributions

The main contributions of this paper include:

- Development of a lightweight EMPO-inspired framework.
- Integration of memory storage and retrieval mechanisms.
- Implementation of novelty-driven exploration rewards.
- Experimental evaluation under limited computational resources.
- Analysis of PPO limitations in sparse-reward environments.

II. BACKGROUND AND RELATED WORK

A. Reinforcement Learning

Reinforcement Learning is based on the interaction between an agent and an environment [2]. The agent learns a policy that maps states to actions in order to maximize cumulative rewards. The RL framework is commonly represented as a Markov Decision Process (MDP) consisting of a state space $\mathcal{S}$, action space $\mathcal{A}$, transition function P , reward function R , and policy π .

B. Proximal Policy Optimization (PPO)

PPO was introduced by Schulman et al. [1] as a stable policy-gradient algorithm. PPO improves training stability by constraining policy updates through a clipped objective function. Its advantages include stable learning, sample efficiency, ease of implementation, and strong empirical performance. Because of these properties, PPO was selected as the baseline reinforcement learning algorithm for this work.

C. Memory-Augmented Reinforcement Learning

Traditional RL methods typically rely on current observations and learned parameters. Memory-augmented approaches extend this paradigm by allowing agents to store and retrieve useful experiences [7]. Memory systems can improve long-term planning, exploration efficiency, knowledge reuse, and sparse-reward learning.

D. EMPO²

EMPO² (Exploratory Memory-Augmented LLM Agent via Hybrid On- and Off-Policy Optimization) [6] introduces a memory-driven exploration framework that combines memory retrieval and reinforcement learning optimization. Key ideas include experience memory, retrieval of useful past experiences, exploration guidance, and hybrid policy optimization.

E. Research Gap

Although EMPO² demonstrates promising results, reproducing the full architecture requires substantial computational resources. This work addresses the gap by proposing EMPO-Light, a simplified framework that preserves the fundamental concepts of memory storage, retrieval, and exploration while remaining computationally feasible on Kaggle and Colab platforms.

III. PROPOSED METHODOLOGY

A. Overview

The proposed framework, EMPO-Light, is a lightweight memory-augmented reinforcement learning architecture inspired by EMPO². The overall architecture consists of four major components: (1) PPO Agent, (2) Experience Memory Buffer, (3) Memory Retrieval System, and (4) Novelty-Based Intrinsic Reward Module.

B. PPO Baseline Agent

Proximal Policy Optimization (PPO) was selected as the baseline reinforcement learning algorithm because of its stability and widespread adoption. The PPO agent was implemented using Stable-Baselines3 [4] and trained on MiniGrid environments [3] with the following hyperparameters: learning rate 3×10^{-4} , discount factor $\gamma = 0.99$, batch size 64, and 2,048 steps per rollout.

C. Experience Memory Buffer

A memory buffer was implemented to store experiences encountered by the agent during interaction with the environment. Each stored experience contains: current state, action, reward, next state, episode termination status, and episode reward. The memory buffer was implemented using a fixed-size deque with a capacity of 5,000 experiences.

D. Memory Retrieval System

Inspired by EMPO², a retrieval mechanism was developed to identify relevant experiences from memory using cosine similarity. For two states $\mathbf{x}$ and $\mathbf{y}$, the similarity is defined as:

$$\text{Similarity}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|} \quad (1)$$

Higher similarity values indicate greater relevance between experiences. The retrieval module supports best-match retrieval and top- k retrieval, where $k = 5$ was used in experiments.

E. Novelty-Based Intrinsic Reward

To encourage exploration in sparse-reward environments, a novelty-based intrinsic reward mechanism was introduced. The novelty module tracks visited states and rewards states that differ significantly from prior experiences. A state is considered novel if its maximum cosine similarity to previously observed states falls below a threshold of 0.95. Novel states receive an intrinsic reward bonus of 0.1.

F. EMPO-Light Framework

The final EMPO-Light architecture combines PPO, the memory buffer, memory retrieval, and the novelty reward module. The framework focuses on the fundamental concepts of experience storage, experience retrieval, and exploration guidance, while remaining computationally feasible for educational and research purposes.

IV. EXPERIMENTAL SETUP

A. Environments

Two MiniGrid benchmark environments were selected for evaluation.

MiniGrid-Empty-8x8-v0 is a simple navigation task in which the agent must observe the environment and navigate toward the goal state. It features a dense reward structure and minimal exploration requirements.

MiniGrid-DoorKey-8x8-v0 is a significantly more challenging environment with a sparse reward structure, multi-stage task completion, long-horizon planning, and significant

exploration requirements. This environment closely reflects the exploration challenges targeted by EMPO².

B. Hardware and Software Configuration

Experiments were conducted on Kaggle Notebooks and Google Colab Pro using Python 3.12, Stable-Baselines3, Gymnasium [5], MiniGrid, NumPy, Pandas, and Matplotlib.

C. Training Configuration

The PPO agent was trained with: learning rate 0.0003, discount factor 0.99, batch size 64, MLP policy, and 50,000+ timesteps for Empty-8x8 and 200,000 timesteps for DoorKey-8x8.

D. Evaluation Metrics

The following metrics were used: average reward, maximum reward, success rate, memory size, novel states found, and top retrieved experience reward.

V. RESULTS AND DISCUSSION

A. PPO Results on MiniGrid-Empty-8x8-v0

The PPO baseline successfully learned the navigation task. As shown in Table I, the agent achieved an average reward of 0.99, validating the correctness of the PPO implementation.

TABLE I
PPO PERFORMANCE ON MINIGRID-EMPTY-8X8-V0

Metric	Value
Average Reward	0.99
Training Timesteps	50,000+
Task Completion	Successful

B. PPO Results on MiniGrid-DoorKey-8x8-v0

The PPO agent failed to solve the DoorKey environment despite 200,000 timesteps of training, as shown in Table II. This result highlights the exploration challenge posed by sparse-reward environments.

TABLE II
PPO PERFORMANCE ON MINIGRID-DOORKEY-8X8-V0

Metric	Value
Average Reward	0.00
Success Rate	0%
Training Timesteps	200,000
Task Completion	Failed

C. EMPO-Light Memory Statistics

The memory system successfully stored and managed exploration experiences. Table III summarises the key memory statistics.

TABLE III
EMPO-LIGHT MEMORY STATISTICS

Metric	Value
Memory Size	5,000
Top Experiences Stored	50
Novel States Found	320
Top Retrieved Experience Reward	63.9

TABLE IV
CONCEPTUAL COMPARISON: PPO VS. EMPO-LIGHT

Feature	PPO	EMPO-Light
Policy Optimization	✓	✓
Experience Memory	×	✓
Memory Retrieval	×	✓
Novelty Tracking	×	✓
Exploration Analysis	Limited	Enhanced

D. Comparison Between PPO and EMPO-Light

Table IV provides a conceptual comparison between standard PPO and the proposed EMPO-Light framework. The proposed framework extends PPO by incorporating mechanisms for experience preservation and retrieval, increasing the agent’s ability to analyze and organize exploration information.

E. Why DoorKey Remains Difficult

Several factors contributed to the failure of PPO in the DoorKey environment. Rewards are sparse and delayed, and the task requires multiple dependent actions. Failure at any stage prevents reward acquisition, providing little useful learning signal during early training. This challenge motivates the development of memory-augmented exploration frameworks such as EMPO².

F. Relationship to EMPO²

EMPO² introduces memory retrieval and exploration guidance mechanisms that enable agents to leverage past experiences during learning. The proposed EMPO-Light framework captures several key ideas from EMPO²: experience storage, experience retrieval, exploration tracking, and memory-driven analysis. However, the complete EMPO² framework additionally includes Large Language Models, hybrid on-policy and off-policy optimization, advanced memory reasoning, and long-horizon planning — components that were beyond the computational scope of this work.

G. Limitations

The memory system does not directly influence PPO policy updates. The retrieval module operates independently from the policy network. Experiments were conducted on relatively small MiniGrid environments, and computational limitations prevented reproduction of the full EMPO² framework.

VI. CONCLUSION AND FUTURE WORK

This paper investigated the feasibility of incorporating memory-augmented exploration mechanisms into reinforcement learning through a lightweight framework inspired by

EMPO². A complete implementation of EMPO-Light was developed using PPO, an experience memory buffer, cosine-similarity-based retrieval, and novelty-based intrinsic rewards.

Experimental results demonstrated that PPO successfully solved MiniGrid-Empty-8x8-v0 with an average reward of 0.99, but failed to solve the more challenging MiniGrid-DoorKey-8x8-v0 after 200,000 training timesteps. The memory system successfully stored 5,000 experiences, identified 320 novel states, and retrieved high-value experiences with a maximum episode reward of 63.9.

Future Work. Several promising directions can extend this work: (1) direct policy integration to allow memory to influence action selection; (2) advanced retrieval mechanisms using neural retrieval models or transformer-based memory systems; (3) curiosity-driven exploration with prediction-error or information-gain rewards; (4) evaluation on larger environments such as Procgen, Atari, and Minecraft; and (5) full EMPO² reproduction incorporating Large Language Models and hybrid optimization.

Memory-augmented reinforcement learning represents an important step toward more capable and autonomous AI agents. The findings of this paper provide a foundation for future research into memory-driven exploration and agentic reinforcement learning systems.

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